

CUSTOMER CHURN ANALYSIS USING PYTHON, SQL AND POWER BI

Company Project Report

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Project Domain

Data Analytics

Technologies Used

Python • SQL • Microsoft Power BI • GitHub

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INTRODUCTION

Customer churn refers to the percentage of customers who discontinue using a company's products or services within a specific period. In highly competitive industries such as telecommunications, customer retention has become one of the most important business objectives because acquiring new customers is significantly more expensive than retaining existing ones.

With the rapid growth of digital services, telecom companies generate large volumes of customer data every day. This data contains valuable information about customer demographics, service subscriptions, billing history, payment methods, contract types, internet services, tenure, and customer behavior. By analyzing this information, organizations can identify the factors that influence customer churn and develop effective customer retention strategies.

The primary objective of this project is to perform customer churn analysis using modern data analytics tools including Python, SQL, and Microsoft Power BI. Python is used for data preprocessing, data cleaning, exploratory data analysis (EDA), and visualization. SQL is utilized to retrieve, filter, aggregate, and analyze customer records efficiently. Microsoft Power BI is used to create interactive dashboards that help business users monitor customer behavior and key performance indicators (KPIs).

The project follows a structured data analytics workflow beginning with data collection, data preprocessing, exploratory analysis, SQL-based business analysis, dashboard development, and business interpretation. Each stage contributes to generating meaningful insights that support data-driven business decisions.

The analysis focuses on identifying customer segments with higher churn probability based on important business attributes such as contract type, tenure, monthly charges, payment method, internet service, and customer demographics. These insights enable organizations to reduce customer attrition, improve customer satisfaction, optimize marketing campaigns, and increase long-term profitability.

Overall, this project demonstrates the practical application of data analytics techniques in solving real-world business problems and highlights how data-driven decision-making can improve customer retention strategies.

BUSINESS PROBLEM

In today's highly competitive telecommunications industry, customer retention has become one of the most significant business challenges. Every customer who discontinues a service represents not only a direct loss of revenue but also an increase in customer acquisition costs, as attracting a new customer is generally more expensive than retaining an existing one.

Telecommunication companies collect large amounts of customer data every day, including demographic information, billing details, service subscriptions, payment methods, internet usage, contract types, and customer tenure. Despite having access to this valuable data, many organizations struggle to effectively identify customers who are at risk of leaving.

Without proper analysis, businesses are unable to understand the reasons behind customer churn, making it difficult to design targeted retention strategies. This can lead to reduced customer satisfaction, declining profitability, and weakened competitive advantage.

The key business problem addressed in this project is to analyze historical customer data and identify the major factors contributing to customer churn. By leveraging Python for data analysis, SQL for business queries, and Microsoft Power BI for interactive visualization, the project aims to generate actionable insights that support better business decision-making and improve customer retention.

The findings of this analysis can help organizations identify high-risk customer segments, optimize marketing efforts, improve customer service, and develop proactive strategies to reduce customer attrition and enhance long-term customer loyalty.

PROJECT OBJECTIVES

The primary objective of this project is to perform customer churn analysis using Python, SQL, and Microsoft Power BI to identify the major factors influencing customer attrition and provide data-driven recommendations for improving customer retention.

The specific objectives of this project are:

- To understand the structure and characteristics of the customer churn dataset.
- To clean and preprocess the dataset by handling missing values and preparing the data for analysis.
- To perform Exploratory Data Analysis (EDA) to identify trends and patterns in customer behavior.
- To use SQL queries for analyzing customer information and generating business insights.
- To develop an interactive Power BI dashboard for effective data visualization.
- To identify customer segments with a higher probability of churn.
- To provide actionable business recommendations that support customer retention strategies.
- To demonstrate the practical application of Python, SQL, and Power BI in solving real-world business problems.

DATASET DESCRIPTION

The dataset used in this project is the **Telco Customer Churn Dataset**, which contains customer demographic information, service details, billing records, contract information, and customer churn status. This dataset is widely used for customer churn analysis and business intelligence projects because it provides comprehensive information required to understand customer behavior and identify factors influencing customer attrition.

The dataset consists of customer records collected from a telecommunication company. Each record represents an individual customer along with various attributes describing their subscription details, payment methods, tenure, monthly charges, total charges, internet services, and churn status.

The dataset was cleaned and preprocessed before analysis to ensure data quality and consistency. Missing values, inconsistent data types, and unnecessary attributes were handled appropriately to improve the accuracy of the analysis.

The cleaned dataset was then used for exploratory data analysis, SQL-based business analysis, and Power BI dashboard development to generate meaningful business insights.

Dataset Information

Attribute	Description
Dataset Name	Telco Customer Churn Dataset
Domain	Telecommunications
File Format	CSV
Number of Records	Approximately 7,043 Customers
Number of Attributes	21 Columns
Target Variable	Churn
Data Type	Structured Data
Tools Used	Python, SQL, Power BI

TECHNOLOGY STACK

The project was developed using multiple data analytics technologies to perform data preprocessing, analysis, visualization, and business reporting.

Technology	Purpose
Python	Data Cleaning, Data Preprocessing, Exploratory Data Analysis
Pandas	Data Manipulation
NumPy	Numerical Computation
Matplotlib	Data Visualization
SQL	Data Querying and Business Analysis
Microsoft Power BI	Interactive Dashboard Development
GitHub	Version Control and Project Management
Microsoft Excel	Initial Data Inspection

Advantages of the Technology Stack

- Python provides efficient libraries for handling large datasets and performing exploratory data analysis.
- SQL enables fast retrieval and aggregation of customer information.
- Power BI creates interactive dashboards that support business decision-making.
- GitHub maintains version control and project documentation.
- Excel helps in quick data verification and validation.

PROJECT METHODOLOGY

1. Data Collection

The customer churn dataset was collected in CSV format. The dataset contains customer demographic information, service details, contract information, billing records, and customer churn status.

2. Data Understanding

The dataset was examined to understand its structure, identify the available attributes, determine data types, and detect missing or inconsistent values before performing any analysis.

3. Data Preprocessing

The collected dataset was cleaned by handling missing values, correcting data types, removing inconsistencies, and preparing the dataset for analysis. Data preprocessing ensures that the analytical results are accurate and reliable.

4. Exploratory Data Analysis (EDA)

Python libraries such as Pandas and Matplotlib were used to explore customer behavior and identify trends related to customer churn. Various charts and statistical summaries were generated to understand relationships among different variables.

5. SQL Analysis

Structured Query Language (SQL) was used to retrieve, filter, group, and summarize customer information. SQL queries helped generate business insights such as customer distribution, churn rate, and contract-wise analysis.

6. Dashboard Development

Microsoft Power BI was used to create an interactive dashboard containing Key Performance Indicators (KPIs), charts, slicers, and visual reports.

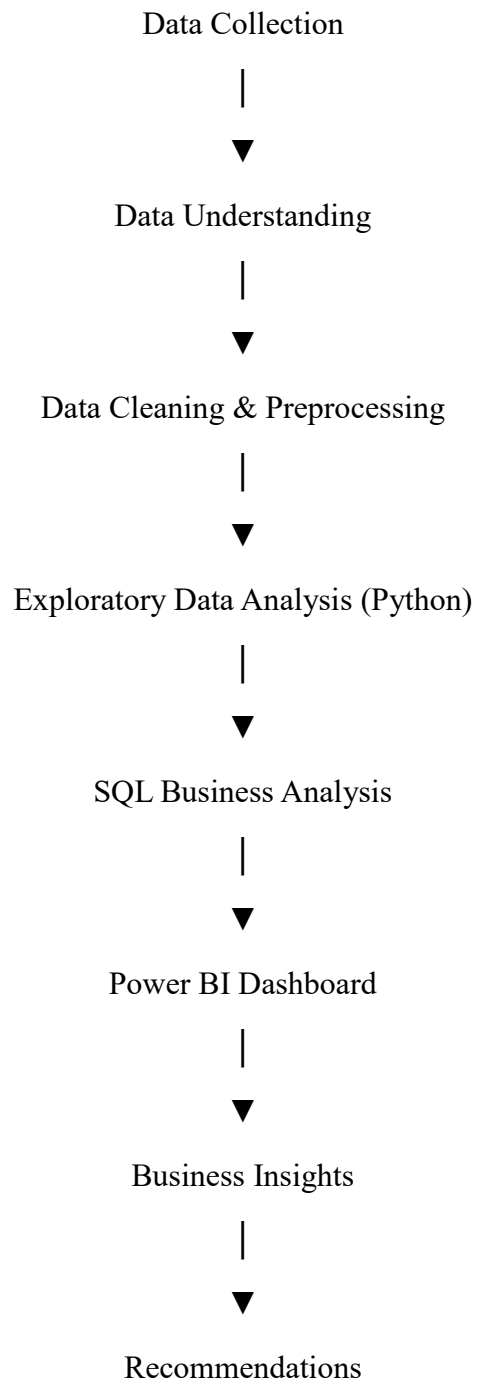
7. Business Interpretation

The final stage involved interpreting the analytical results and providing practical recommendations to improve customer retention and support business decision-making.

PROJECT WORKFLOW

Stage	Description
Data Collection	Collect the telecom customer churn dataset.
Data Understanding	Analyze dataset structure and attributes.
Data Cleaning	Handle missing values and prepare clean data.
Exploratory Data Analysis	Analyze customer behavior using Python.
SQL Analysis	Generate business insights using SQL queries.
Dashboard Development	Build interactive Power BI dashboards.
Business Insights	Interpret results and identify churn patterns.
Recommendations	Suggest strategies to reduce customer churn.

Workflow Diagram



DATA PREPROCESSING

Data preprocessing is one of the most important stages in any data analytics project. Raw datasets often contain missing values, inconsistent formatting, duplicate records, and incorrect data types that can negatively impact the quality of analysis. Therefore, preprocessing was performed before conducting exploratory data analysis and visualization.

The Telco Customer Churn dataset was imported into Python using the Pandas library. The structure of the dataset was examined to identify missing values, data types, and potential inconsistencies. Each column was reviewed to ensure the data was complete and suitable for analysis.

The preprocessing activities performed in this project include:

- Importing the dataset into Python.
- Identifying missing and null values.
- Removing duplicate records.
- Converting data types where required.
- Cleaning inconsistent values.
- Preparing the dataset for analysis.
- Exporting the cleaned dataset for SQL and Power BI.

These preprocessing steps improved data quality and ensured that the analysis produced reliable and accurate business insights.

DATA CLEANING PROCESS

Cleaning Activity	Purpose
Import Dataset	Load customer data into Python
Check Missing Values	Identify incomplete records
Remove Duplicates	Improve data quality
Convert Data Types	Ensure consistency
Handle Null Values	Prevent analysis errors
Export Clean Dataset	Prepare data for SQL & Power BI

The cleaned dataset was saved as:

cleaned_customer_churn.csv

This dataset was used throughout the remaining phases of the project.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is the process of analyzing data using statistical summaries and visualizations to discover meaningful patterns, trends, and relationships. Python libraries such as Pandas and Matplotlib were used to perform the analysis.

The analysis focused on understanding customer behavior across different business attributes including:

- Customer Churn Distribution
- Gender Distribution
- Contract Types
- Internet Services
- Payment Methods
- Monthly Charges
- Total Charges
- Customer Tenure
- Senior Citizen Distribution

The visualizations generated during EDA helped identify the customer groups that were more likely to discontinue their services.

EDA OBSERVATIONS

Based on the exploratory analysis, the following observations were identified:

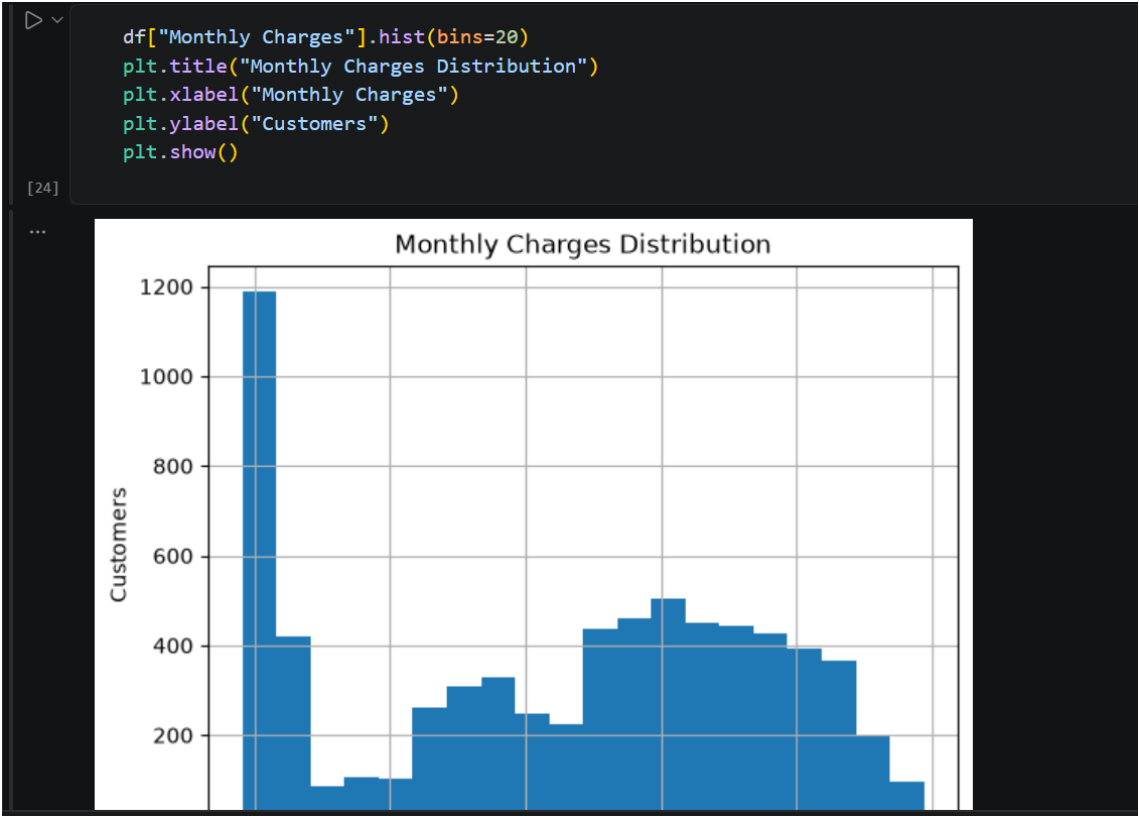
- Customers with month-to-month contracts showed a higher churn rate.
- Customers with shorter tenure were more likely to leave the service.
- Higher monthly charges were associated with increased churn.
- Customers using electronic payment methods showed comparatively higher churn.
- Long-term contract customers demonstrated better retention.
- Internet service type had a measurable impact on customer churn.
- Customer tenure played a significant role in predicting churn behavior.

These observations provide valuable business insights for designing customer retention strategies.

EDA VISUALIZATIONSEDA Visualizations

The following visualizations were created using Python to understand customer behavior:

Visualization	Purpose
Churn Distribution	Analyze customer attrition
Contract Type Analysis	Compare churn by contract
Monthly Charges Distribution	Study customer billing patterns
Tenure Analysis	Understand customer loyalty
Internet Service Analysis	Evaluate service preferences
Payment Method Analysis	Identify payment trends
Gender Distribution	Compare male and female customers



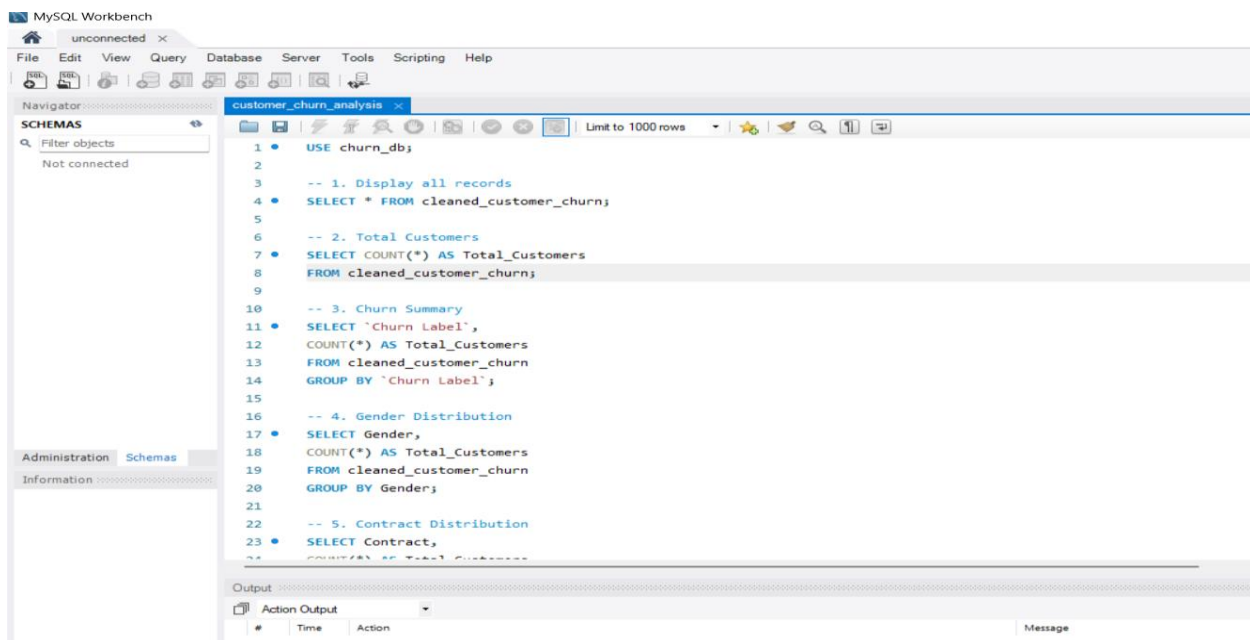
SQL ANALYSIS

Structured Query Language (SQL) was used to retrieve, filter, aggregate, and analyze customer data stored in the cleaned dataset. SQL enabled efficient extraction of business insights by grouping, summarizing, and filtering customer information based on different business attributes.

The SQL operations performed in this project include:

- Data Selection
- Data Filtering
- Sorting Records
- Aggregate Functions
- Group By Analysis
- Conditional Queries
- Customer Segmentation
- Business Summary Reports

SQL helped transform raw customer data into structured business information that was later visualized in Microsoft Power BI.



SQL QUERIES IMPLEMENTED

The following SQL queries were executed during the project:

Query	Purpose
SELECT	Retrieve customer records
WHERE	Filter customer data
GROUP BY	Analyze customer categories
ORDER BY	Sort customer information
COUNT()	Count total customers
AVG()	Calculate average monthly charges
SUM()	Summarize revenue-related values
MAX()	Identify highest values
MIN()	Identify lowest values

SQL INSIGHTS

The SQL analysis generated several important business insights regarding customer churn.

The analysis identified that:

- Month-to-month contract customers contributed significantly to customer churn.
- Customers with shorter tenure showed higher attrition rates.
- Electronic payment methods were commonly associated with churned customers.
- Long-term contract customers demonstrated better customer retention.
- Higher monthly charges increased the probability of customer churn.

These findings provide valuable support for business managers in designing customer retention programs.

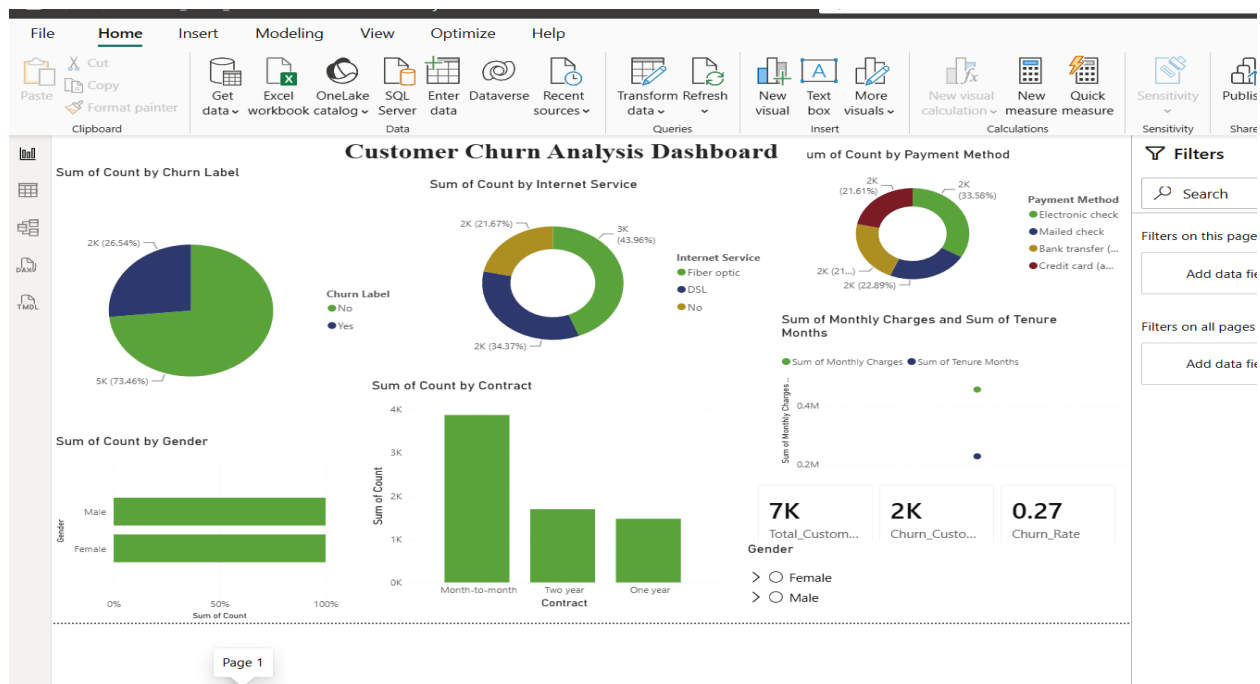
POWER BI DASHBOARD

Microsoft Power BI was used to create an interactive dashboard that presents customer churn information through dynamic charts, KPI cards, filters, and visual reports.

The dashboard enables users to monitor customer behavior, compare churn across different categories, and identify important business trends.

The dashboard contains the following components:

Dashboard Component	Purpose
KPI Cards	Display important business metrics
Bar Charts	Compare customer categories
Pie Charts	Show customer distribution
Line Charts	Analyze trends
Filters (Slicers)	Enable interactive analysis
Tables	Display summarized business data



KEY PERFORMANCE INDICATORS (KPIs)

The Power BI dashboard presents several Key Performance Indicators (KPIs) that help organizations evaluate customer retention and business performance.

The major KPIs included in the dashboard are:

KPI	Business Importance
Total Customers	Overall customer base
Total Churn Customers	Customers who discontinued services
Active Customers	Customers currently using services
Churn Rate	Percentage of customers lost
Average Monthly Charges	Revenue analysis
Average Customer Tenure	Customer loyalty indicator
Contract Distribution	Customer subscription analysis

KEY BUSINESS INSIGHTS

The analysis of the customer churn dataset revealed several important business insights that can help organizations improve customer retention strategies.

Insight 1: Contract Type

Customers with **Month-to-Month contracts** showed the highest churn rate compared to One-Year and Two-Year contract customers.

Insight 2: Customer Tenure

Customers with shorter service tenure were more likely to discontinue their services, whereas long-term customers demonstrated higher loyalty.

Insight 3: Monthly Charges

Customers paying higher monthly charges showed a greater tendency to churn, indicating that pricing plays a significant role in customer retention.

Insight 4: Payment Method

Customers using Electronic Check as their payment method exhibited a comparatively higher churn rate than customers using automatic payment methods.

Insight 5: Internet Services

Internet service subscriptions influenced customer churn patterns, suggesting that service quality and customer satisfaction are important factors affecting customer retention.

Overall, the analysis demonstrates that customer churn is influenced by multiple business variables, and organizations can reduce churn through targeted retention strategies.

BUSINESS RECOMMENDATIONS

Based on the analytical findings, the following recommendations are proposed to improve customer retention and reduce customer churn.

- Introduce attractive discounts for customers choosing long-term contracts.
- Strengthen customer support services for new customers during the first six months.
- Review pricing strategies for customers with higher monthly charges.
- Promote automatic payment methods to improve customer convenience.
- Implement customer loyalty and reward programs.
- Conduct regular customer satisfaction surveys.
- Use predictive analytics models to identify high-risk customers before churn occurs.
- Develop personalized retention campaigns for customers with high churn probability.

Implementing these recommendations can significantly improve customer satisfaction, increase customer loyalty, and reduce revenue loss caused by customer churn.

CHALLENGES FACED

During the development of this project, several technical and analytical challenges were encountered.

- Understanding the dataset structure and business context.
- Handling missing and inconsistent data.
- Performing effective data preprocessing.
- Designing meaningful visualizations.
- Writing optimized SQL queries.
- Creating an interactive Power BI dashboard.
- Interpreting analytical results into business recommendations.

These challenges were successfully addressed through systematic data analysis and continuous validation of results.

CONCLUSION

The **Customer Churn Analysis Using Python, SQL, and Microsoft Power BI** project successfully analyzed customer behavior and identified the major factors contributing to customer churn.

Python was used for data cleaning, preprocessing, and exploratory analysis, SQL was utilized for business data analysis, and Microsoft Power BI was employed to build an interactive dashboard for visualization and reporting.

The project demonstrated that customers with month-to-month contracts, shorter tenure, and higher monthly charges are more likely to discontinue their services.

The developed dashboard enables business users to monitor customer behavior, identify high-risk customer groups, and make informed decisions to improve customer retention.

Overall, this project demonstrates the importance of data analytics in solving real-world business problems and highlights how modern analytical tools can transform raw data into valuable business intelligence.

FUTURE SCOPE

This project can be further enhanced by incorporating advanced analytical and machine learning techniques.

Future improvements include:

- Developing machine learning models for customer churn prediction.
- Deploying predictive models using cloud platforms.
- Automating dashboard refresh using Power BI Service.
- Integrating real-time customer data.
- Building a web application for churn prediction.
- Applying Artificial Intelligence for personalized customer recommendations.

These enhancements will improve prediction accuracy and support proactive customer retention strategies.

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5. Matplotlib Documentation — <https://matplotlib.org/>
6. Microsoft Power BI Documentation — <https://learn.microsoft.com/power-bi/>
7. SQL Documentation — <https://www.w3schools.com/sql/>
8. GitHub Documentation — <https://docs.github.com/>

GITHUB REPOSITORY

The complete source code, dataset, SQL scripts, Power BI dashboard, project documentation, and supporting files for this project are available in the following GitHub repository:

Repository Name:

Customer-Churn-Analysis

GitHub Repository Link:

<https://github.com/rajikumar9894-hue/Customer-Churn-Analysis>

The repository includes:

- Python Notebook
- SQL Scripts
- Cleaned Dataset
- Power BI Dashboard
- Project Documentation
- Screenshots
- README File
- License
- Requirements File

APPENDIX – PYTHON IMPLEMENTATION

Python was used for data preprocessing, exploratory data analysis (EDA), and visualization. The implementation involved importing the dataset, cleaning missing values, converting data types, and generating analytical charts to understand customer churn behavior.

The following Python libraries were used during the project:

Library	Purpose
Pandas	Data manipulation and analysis
NumPy	Numerical computations
Matplotlib	Data visualization
OS	File handling

The Python notebook contains the complete implementation of:

- Dataset Import
- Data Cleaning
- Missing Value Handling
- Exploratory Data Analysis
- Data Visualization
- Exporting Clean Dataset

APPENDIX – SQL IMPLEMENTATION

SQL was used to analyze customer data and generate business insights.

The SQL operations performed include:

- Data Selection
- Data Filtering
- Aggregate Functions
- Group By Analysis
- Sorting Records
- Business Queries

Sample SQL Query

```
SELECT Contract, COUNT(*) AS Total_Customers  
FROM customer_churn  
GROUP BY Contract;
```

APPENDIX – POWER BI DASHBOARD

Microsoft Power BI was used to build an interactive dashboard for customer churn analysis.

The dashboard includes:

- KPI Cards
- Customer Distribution
- Churn Analysis
- Contract Type Analysis
- Payment Method Analysis
- Internet Service Analysis
- Monthly Charges Analysis
- Interactive Filters (Slicers)

The dashboard enables management to monitor customer behavior and make informed business decisions.

PROJECT OUTCOME

The Customer Churn Analysis project successfully demonstrated the practical application of data analytics techniques in solving a real-world business problem. By integrating Python, SQL, and Microsoft Power BI, the project transformed raw customer data into meaningful business insights.

The project achieved all its planned objectives, including data preprocessing, exploratory data analysis, SQL-based business analysis, dashboard development, and business recommendation generation. The findings can support organizations in improving customer retention strategies and reducing customer churn.

This project enhanced technical skills in Python programming, SQL querying, data visualization, dashboard development, and business analytics while providing practical experience in handling production-level business data.

